STUDY NOTES — ARTIFICIAL INTELLIGENCE

Natural Language Processing

From Classical NLP foundations to text processing pipelines, challenges, and core NLP tasks — organized with examples.

1 What is NLP?

Natural Language Processing (NLP) = the field of AI that enables computers to understand, interpret, and generate human language. It sits at the intersection of **Computational Linguistics** and **Machine Learning**.

THREE CORE CAPABILITIES

Understanding → Interpretation → Generation

SIMPLE DEFINITION
NLP is how machines read and write human language. When you ask Siri a question, NLP understands your words. When it replies, NLP generates the response.

PHASE 1 — CLASSICAL NLP
Rule-based systems, hand-crafted grammar rules, statistical models. Relied on linguistic knowledge engineered by humans.

Tools: Bag of Words, TF-IDF, N-grams, POS taggers, regex parsers.

PHASE 2 — DEEP LEARNING NLP
Neural networks learn language patterns automatically from data. Context is understood at a much deeper level.

Tools: Word2Vec, BERT, GPT, Transformers, LLMs.

2 Challenges in NLP

① Ambiguity — Words can have different meanings

The same word can mean completely different things depending on usage. This is one of the hardest problems in NLP.

SAME WORD, DIFFERENT MEANINGS

Bank → Financial Institution
Bank → River bank / side of a river

Throne → King's Throne (seat of power)
Throne → "Roses have thorns" (homophone confusion)

WHY THIS IS HARD

Without context, a machine cannot decide which meaning applies. Even humans sometimes need a full sentence to disambiguate.

EXAMPLE

"I went to the bank." — Did you deposit money or sit by

② Context Dependency — Meaning depends on surrounding words

Words and pronouns derive meaning from the broader sentence or paragraph, not in isolation.

CONTEXT DEPENDENCY EXAMPLE

"**Paul** is a good boy having good **morals**." — here "good" refers to moral character.

"**Paul** is good in playing **cricket**." — here "good" refers to athletic skill.

"I want to meet **John** but **he** was sleeping."

→ The pronoun **"he"** refers back to John — a machine must track this dependency across words. This is called **co-reference resolution**.

③ Complex Grammar — Rules, Sarcasm & POS

Natural languages have highly irregular grammar rules, idioms, and nuances like **sarcasm** that don't follow any rule.

PARTS OF SPEECH (POS) — ENGLISH HAS 8

Noun

Names a person, place, or thing.

e.g. Paul, Bangkok, river

Pronoun

Replaces a noun.

e.g. he, she, they, it

Verb

Describes an action or state.

e.g. run, is, played

Adverb

Modifies a verb, adjective, or other adverb.

e.g. quickly, very, well

Adjective

Describes a noun.

e.g. good, tall, fast

Preposition

Shows relationship between words.

e.g. in, at, on, by

Conjunction

Connects words or clauses.

e.g. and, but, or, yet

Interjection

Expresses emotion; stands alone.

e.g. Oh!, Wow!, Hey!

④ Ever-Growing Vocabulary — New words emerge constantly

Language evolves — new slang, technical terms, brand names, and abbreviations appear constantly. Classical NLP systems struggle with out-of-vocabulary (OOV) words.

REAL-WORLD IMPACT
Words like "selfie", "bitcoin", "COVID", "GPT" didn't exist in older training data. A model trained before these terms will fail to understand or generate them properly.

3 Classical NLP Pipeline

NLP deals with **textual data**. But computers only understand numbers. So the pipeline converts raw text into numeric representations that models can process.

DATA TYPES

Numbers = 1, 2, 3 **Vectors** = [1.01, 1.02, ...] **Tensors** = Vectors processed on GPU

PIPELINE FLOW

Raw Text → Text Preprocessing → Text Representation → NLP Task

4 Text Preprocessing

Before feeding text to a model, it must be cleaned and normalized. The four key steps:

① Tokenization

Breaking text into smaller units (tokens) — words, subwords, or characters.

EXAMPLE
"I love NLP" → ["I", "love", "NLP"] (word-level tokenization) "playing" → ["play", "##ing"] (subword tokenization used in BERT) Note: Tokenization can sometimes cause **loss of information** if not done carefully.

② Stemming & Lemmatization

Reducing words to their base/root form to treat variations as the same word.

STEMMING (CRUDE)

Chops off word endings. Fast but sometimes inaccurate. May not produce real words.

```
"playing" → "play"
"studies" → "studi" (not a real word!)
```

LEMMATIZATION (ACCURATE)

Uses vocabulary and grammar to find the true root word. Slower but always produces valid words.

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"better" → "good"
"best" → "good"
"played" → "play"
```

③ Stopwords Removal

Common words that carry little meaning are removed to reduce noise and focus on key words.

COMMON STOPWORDS

"in", "at", "the", "is", "a", "an", "and", "or", "but", "of", "to"

BEFORE & AFTER

Before: "The cat is sitting on the mat" After: "cat sitting mat" → cleaner signal for the model

④ Normalization

Standardizing text format so equivalent words are treated identically.

CASE NORMALIZATION

Problem: "Apple", "apple", "APPLE" are the same word but look different to a machine.

Solution: Convert everything to **lowercase** → "apple", "apple", "apple"

Other normalizations: removing punctuation, expanding contractions ("don't" → "do not"), handling numbers.

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5 Text Representation (Embeddings)

After preprocessing, text must be converted into **numerical vectors** that capture meaning. This is called embedding.

i One-Hot Encoding (OHE)

Each word is represented as a binary vector with 1 at its position and 0 everywhere else.

EXAMPLE

"cat" in a 5-word vocab → [1,0,0,0,0]. Simple but huge vectors with no semantic meaning — "cat" and "kitten" look completely unrelated.

ii Bag of Words (BoW)

Represents a document as a count of how many times each word appears. Order is ignored.

EXAMPLE

"I love cats. Cats are great." → {"I":1, "love":1, "cats":2, "are":1, "great":1}. Loses word order and context.

iii TF-IDF

Term Frequency × Inverse Document Frequency. Words common across all docs get lower weight; rare, important words get higher weight.

EXAMPLE

The word "the" appears in every doc → low TF-IDF score. The word "quantum" in a physics paper → high score. Much better than plain BoW.

iv N-grams

Instead of single words, uses sequences of N consecutive words as features. Captures some local context.

EXAMPLE

Bigrams (N=2): "I love" "love cats" "cats are". Trigrams (N=3): "I love cats" "love cats are". Richer than BoW but vocabulary explodes.

6 Core NLP Tasks

With preprocessed and represented text, NLP models can perform a wide range of downstream tasks:

i POS Tagging — Parts of Speech Tagging

Labeling each word in a sentence with its grammatical role (noun, verb, adjective, etc.).

POS TAGGING EXAMPLE

```
"Paul  [NOUN]  is  [VERB]  living  [VERB]  in  [PREP]  Bangkok  [NOUN]"
```

POS tagging is foundational — it helps downstream tasks like NER, parsing, and machine translation understand sentence structure.

ii NER — Named Entity Recognition

Identifying and classifying named entities in text — people, places, organizations, dates, etc.

NER EXAMPLE

"**Paul** [PERSON] is living in **Bangkok** [LOCATION]"

Entity types include: PERSON, LOCATION, ORGANIZATION, DATE, TIME, MONEY, PRODUCT.

REAL-WORLD USE

News article analysis: "Apple [ORG] CEO Tim Cook [PERSON] announced in Cupertino [LOC] that..." — NER extrac structured facts from unstructured text automatically.

FULL NLP PIPELINE — QUICK REFERENCE

Raw Text
Any textual data: articles, conversations, documents, social media posts.

Preprocessing
Tokenization → Stemming/Lemmatization → Stopword Removal → Normalization

Text Representation
OHE → Bag of Words → TF-IDF → N-grams → (Deep: Word2Vec, BERT embeddings)

NLP Tasks
POS Tagging, NER, Sentiment Analysis, Machine Translation, Summarization, QA
